

Inference Research Ideas — Repository Overview

Inference Overlay Research

Multi-Cloud Inference Benchmarking & Optimization Platform

A research initiative evaluating production LLM inference across 9 cloud providers — 4 hyperscalers + 5 neo-clouds.

Research Vision

Modern LLM inference has moved from single-GPU prototypes to multi-cloud production systems, yet the ecosystem lacks standardized benchmarks, cross-cloud normalization, and quantization certification for regulated industries. This initiative addresses these gaps through four complementary research directions.

Research Directions

#	Project	Key Takeaway	Venue
1	InferMark Benchmarking	3.2× throughput variance across clouds; 18-61% production overhead nobody measures	OSDI

#	Project	Key Takeaway	Venue
2	Disaggregated Cross-Cloud Inference	Multi-hop prefill/decode saves 15-29% cost; GQA/MLA are natural fit for cross-cloud disagg	OSDI/ NSDI
3	Cross-Cloud Performance Parity	4-layer normalization: $\pm 17\%$ $\rightarrow$ $\pm 4\%$ variance; network topology dominates, not GPU compute	SysML
4	Accuracy-Preserving Quantization	SSMs most resilient to quantization; MLA latent KV most sensitive; QJL catches 99.97% errors	NeurIPS
—	Full Bucket List (14 ideas)	Stack-ranked research ideas with takeaways, novelty, and venue fit	—

Cloud Providers Under Evaluation

Hyperscalers

Provider	GPU SKUs	Interconnect	Regions
AWS	H100 SXM (p5), B200 (p5e)	EFA v2 (SRD)	us-east-1, eu-west-1
GCP	H100 (A3 Mega), B200 (A3 Ultra)	GPUDirect-TCPX	us-central1, europe-west4
Azure	H100 (ND v5), B200 (ND v6)	InfiniBand NDR	eastus, westeurope
OCI	B200 bare-metal (BM.GPU.B200.8)	RDMA Cluster Net v2	us-ashburn, eu-frankfurt

Neo-Clouds

Provider	GPU SKUs	Interconnect	Key Differentiator
CoreWeave	H100 SXM, B200	InfiniBand NDR	GPU-native, K8s-first, fastest availability
Lambda	H100 SXM, B200	InfiniBand HDR/ NDR	Simplest pricing, ML-focused
Together AI	H100, custom	Custom fabric	Inference-optimized, open-source hosting
Crusoe Energy	H100 SXM, B200	InfiniBand NDR	Renewable/flare-gas powered, lowest carbon
Voltage Park	H100 SXM	InfiniBand NDR	Large contiguous clusters, HPC-grade

Repository Structure

```
inference-overlay-research/
├─ README.md                                # This file
├─ research-ideas-bucket-list.md            # 14 ideas stack-ranked
with takeaways
├─ 01-infermark-benchmarking/
│   └─ README.md                            # 14-axis benchmarking
framework
│   └─ architecture.drawio                  # System architecture
diagram
├─ 02-disaggregated-cross-cloud/
│   └─ README.md                            # Cross-cloud prefill/
decode optimization
│   └─ architecture.drawio                  # Disaggregation flow
diagram
├─ 03-cross-cloud-parity/
│   └─ README.md                            # 4-layer normalization
framework
│   └─ architecture.drawio                  # Overlay architecture
diagram
└─ 04-accuracy-preserving-quantization/
    └─ README.md                            # TurboQuant
certification framework
    └─ architecture.drawio                  # Evaluation pipeline
diagram
```

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